INCH-POUND

MIL-DTL-17/208A
w/AMENDMENT 1
28 June 2019
SUPERSEDING
MIL-DTL-17/208A
12 June 2014

DETAIL SPECIFICATION SHEET

CABLE, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 185 OHMS,
M17/208-00001

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-DTL-17.

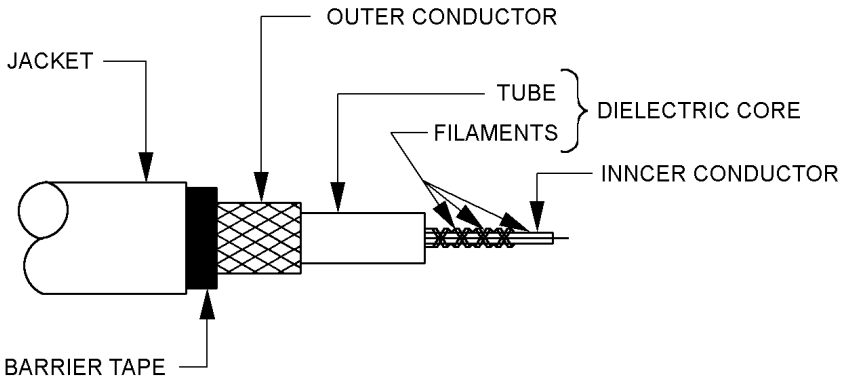


FIGURE 1. General configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Solid silver coated copper wire. Overall diameter: .007 inch ± .0010.
Dielectric core	Type A-3: Air-spaced polyethylene. A braid of 24 polyethylene filament threads, .020 inch approximate diameter each, with 2 picks/inch approximate, under an extruded tube. Overall diameter: .285 inch ± .010. Alternate Type A-3: Air-spaced polyethylene: A braid of 8 polyethylene filament threads, .035 inch approximate diameter each, with 2 picks/inch approximate, under an extruded tube. Overall diameter: .285 inch ± .010.
Outer conductor	Single braid of AWG#34, bare, copper wire. Diameter: 340 inch ± .340. Coverage: 91.1% Carriers: 24 Ends: 8 Picks/inch: 7.0 ± 10%

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TABLE I. Description – Continued.

Components	Construction details
Barrier tape	A .001 inch thick polyester tape faced with a .002 inch thick layer of aluminum. The tape will be applied with a 50% lap, minimum. Aluminum face toward the outer conductor. Diameter: .350 inch, maximum.
Jacket	Type XIV, Cross-linked polyolefin. Diameter: .405 inch ± .010.

ENGINEERING INFORMATION:

Continuous working voltage: 1,000 V rms, maximum.

Operating frequency: 1 GHz, maximum.

Velocity of propagation: 88 percent, nominal.

Power ratings: See figure 2.

Operating temperature range: -40°C to +80°C.

Inner conductor properties:

DC resistance (maximum at +20°C): 54.4 ohms per 100 feet, maximum.

Elongation: 2 percent, minimum.

Tensile strength: 110 klbf/inch/inch, minimum.

Engineering notes: This cable is useful in low capacitance, medium low temperature applications.
(See connector series “N”, “C”, and “SC” per MIL-PRF-39012).
Use this cable in new designs in –lieu-of MIL-C-17/47 cables.

REQUIREMENTS (See table II)

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Diameter: Applicable.

Out-of-roundness: Not applicable.

Eccentricity: 10 percent, maximum.

Adhesion of conductors: Not applicable.

Electrical and mechanical:

Continuity: Applicable.

Spark test: 5,000 V rms, minimum.

Voltage withstanding: 5,000 V rms, minimum.

Insulation resistance: Not applicable.

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Corona extinction voltage: Not applicable.

Characteristic impedance: 185 ± 10.

RF transmission loss (attenuation): 8.5 dB per 100 feet, maximum at .4 GHz.

Standing wave ratio (return loss): Not applicable.

Capacitance: 7.2 pF per foot, maximum.

Capacitance stability: ± 5 percent.

Capacitance unbalance:	Not applicable.
Transmission unbalance:	Not applicable.
Inductance:	Not applicable
Mechanically induced noise:	Not applicable.
Time delay:	Not applicable.

Aging stability: +98° ± 2°C.

Stress crack resistance:	Not applicable.
Outer conductor integrity:	Not applicable.

Cold bend: -40°C ± 2°C.

Dimensional stability:	Not applicable.
Contamination:	Not applicable.
Bendability:	Not applicable.
Flammability:	Not applicable.

Flame propagation: Applicable.

Acid gas generation: 2.0 percent, maximum.

Halogen content: 0.2 percent, maximum.

Immersion test:

Tensile strength, percent of unaged minimum: 50.

Elongation, percent of unaged minimum: 50.

Smoke index: 25 maximum.

Toxicity index: 5 maximum.

Durometer hardness: (Type A) 80 minimum.

Weathering: Applicable.

Abrasion resistance: 75 cycles, minimum (jacket only).

Tear strength: 35 pounds per inch minimum.

Heat distortion: 30 percent maximum distortion.

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Physical tests on unaged jacket:

Tensile strength: 1,300 psi, minimum.

Elongation: 160 percent minimum.

Physical tests on aged jacket:

Air oven:

Tensile strength, percent minimum: 60.

Elongatin, percent minimum: 60.

Hot oil immersion:

Tensile strength, percent minimum: 50.

Elongation, percent minimum: 50.

Tensile strength and elongation: 1,300 psi, 160 percent minimum.

Weight: 8.9 pounds per 100 feet maximum.

Marking: Applicable.

Workmanship: Applicable.

Part or Identifying Number (PIN): M17/208-00001.

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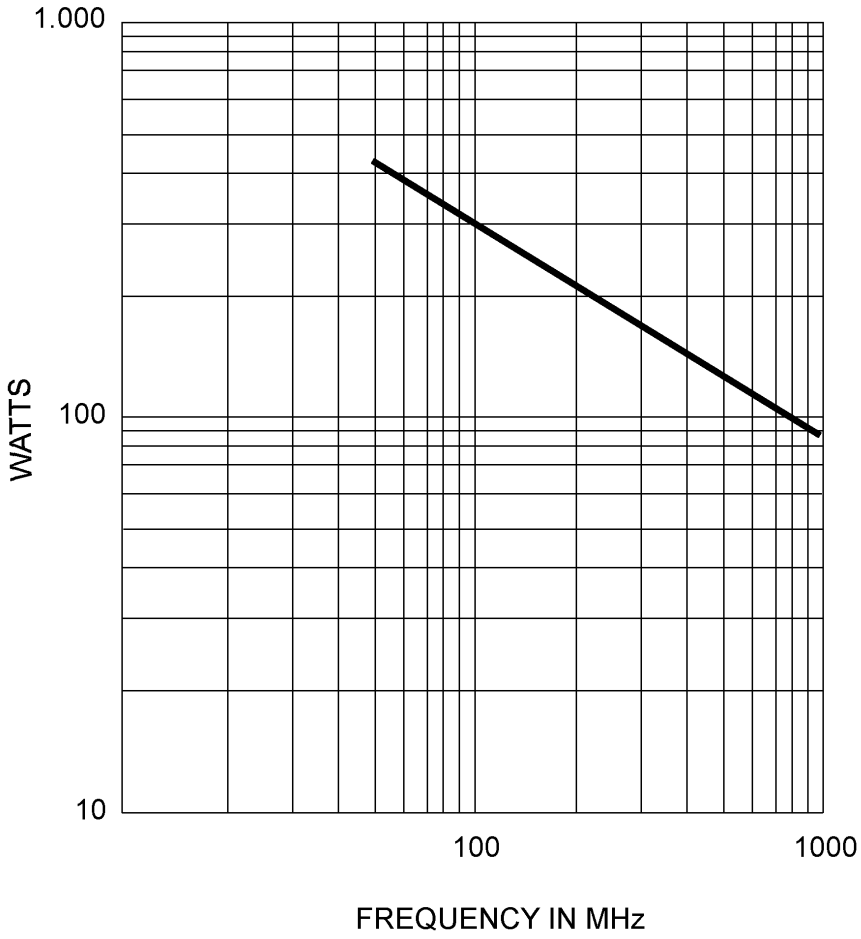


FIGURE 2. Power rating at 25°C sea level.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	N/A				
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	N/A				
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	3.7.10	4.8.11	1	Per Table VIII	C
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				

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TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Time delay	N/A				
	Aging stability	3.7.15	4.8.16	4	Per Table VIII	C
	Stress crack resistance	N/A				
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	N/A				
	Contamination	N/A				
	Bendability	N/A				
	Flammability	N/A				
	Flame propagation	3.7.23	4.8.25	1		
	Acid gas generation	3.7.24	4.8.26	1		
	Halogen content	3.7.25	4.8.27	1		
	Immersion tests	3.7.26	4.8.28	1		
	Smoke index	3.7.27	4.8.29	1		
	Toxicity Index	3.7.28	4.8.30	1		
	Durometer hardness	3.7.29	4.8.31	1		
	Weathering	3.7.30	4.8.32	1		
	Abrasion resistance	3.7.31	4.8.33	5		
	Tear strength	3.7.32	4.8.34	6	Per Table VIII	B
	Heat distortion	3.7.33	4.8.35	1	Per Table VIII	C
	Physicals (aged)	3.7.34	4.8.36	1	Per Table VIII	C
	Tensile strength and elongation	3.7.35	4.8.37	1	Per Table VIII	C
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Amendment notations: The margins of this specification are marked with vertical lines to indicate where modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-17 this document references:

MIL-PRF-39012 MIL-C-17/47

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – AR, AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

(Project 6145-2019-027)

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